Hierarchical view-Myelin NEP readiness signal spine

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Foundational Philosophy (FP 1–5)
 |
 ├── Orientation Signal
 |    └── What educators naturally value and attend to
 |
 ├── Practice Depth Signal (D1–D4)
 |    ├── Observable classroom enactment
 |    ├── FP-specific observability constraints
 |    └── Depth ≠ quality, depth = maturity of translation
 |
 ├── Practice Diagnostic Signal (Areas A1–A4)
 |    ├── Conditions shaping translation
 |    ├── Captured via Agree–Disagree (with sub-areas)
 |    └── Non-judgemental, explanatory
 |
 ├── Behavioural Practice Signal (Micro-tasks, 21 days)
 |    ├── Chosen by teacher (3–5 tasks)
 |    ├── Tasks span 2–3 FPs
 |    ├── Daily logging → consistency, persistence
 |    └── Detects transfer or shift in depth
 |
 └── Cognitive Capacity Signal (MathTangle)
      ├── Apply
      ├── Analyse
      └── Evaluate
```

This hierarchy aligns with SRI's "signals → synthesis → trajectory" logic.

Table 1: Foundational philosophy signal design

FP	Orientation Signal Captures	Practice Depth Observable	Non-Observable / Design Notes
FP-1 **Every Child is Unique**	Belief in learner diversity, inclusion, differentiation	D2–D3 observable	D1 (implicit belief) and D4 (adaptive redesign) not directly observable
FP-2 **Holistic & Experiential**	Learning beyond rote, context, meaning	D1–D3 observable	D4 not directly observable (emerges via transfer)
FP-3 **Reflective Practitioner**	Value of reflection, professional judgement	D1–D4 observable	Often cognitively present before visible
FP-4 **Assessment for Learning**	Belief in formative, diagnostic assessment	D1–D4 observable	Structurally constrained by system
FP-5 **Collaboration & Community**	Openness to collaboration	D1–D4 observable	Highly system-dependent

FP-1 non-observables are *intentionally surfaced* through FP-2 and FP-4 behaviours by design.

Table 2: Practice depth definition (Common Across FPs)

Depth	Meaning (Myelin)
D1	Procedural / compliance-based enactment
D2	Responsive, situational adjustments
D3	Diagnostic, intentional practice
D4	Adaptive redesign based on learner/system signals

Depth is **not a score**; it is a *translation maturity marker*.

Table 3: Practice diagnostics (Teacher Lens)

Area	Purpose	Key Sub-Areas	Response Scale
A1 **Continuous Learning Diagnostics**	Can teachers see learning clearly enough to act?	Access, Usability, Follow-through, Collective Use, Student Clarity	SA / A / D / SD
A2 **Teacher Growth & Safety**	Is change safe and supported?	Structure, Safety, Peer Learning, Experimentation, Growth Culture	SA / A / D / SD
A3 **Holistic Progress Card Enablement**	Is holistic intent operationally shared?	Understanding, Clarity, Time, Use, Student Role	SA / A / D / SD
A4 **Parent & Community Support**	Is collaboration enabled or individualised?	Communication, Guidance, Facilitation, Recognition	SA / A / D / SD

These **explain depth**, they do not define it.

Table 4: Micro-Task behavioural signal (21 Days)

Aspect	Design Choice
Task count	10–12 available
Selection	Teacher chooses 3–5
FP coverage	Each task spans 2–3 FPs
Duration	21 days
Logging	Daily self-logged execution
Signals derived	Consistency, persistence, transfer
Purpose	Detect *behavioural translation* and depth movement

Critical:
Micro-tasks convert *belief into habit* and make **D4 observable over time**.

Table 5: Cognitive capacity signal (MathTangle)

Cognitive Level	What It Captures	Why It Matters
Apply	Ability to use concepts	Baseline execution capacity
Analyse	Ability to decompose, reason	Diagnostic thinking

Evaluate	Ability to judge, adapt	Required for D3–D4 practice

MathTangle is **orthogonal** to FP orientation:

- It explains *why* some teachers plateau at D2
- It strengthens interpretation of practice depth shifts

Table 6: Signals stay on SRI

Principle	How Your Design Respects It
No single instrument dominates	Multiple independent signals
Orientation ≠ readiness	Depth + behaviour required
Conditions explain gaps	A-areas & B-areas
Trajectory over scores	Micro-tasks + logging
Non-evaluative	No rankings, no grades

Myelin's NEP readiness design captures what educators value (orientation), how that value translates into classroom practice (depth), what conditions shape that translation (diagnostics), how behaviour stabilises over time (micro-tasks), and whether cognitive capacity supports deeper change (MathTangle).

myelin's cognitive capacity

Current cognitive signal stack is:

- MathTangle → Apply, Analyse, Evaluate
- Practice Depth (D1–D4) → enactment maturity
- Micro-task consistency (21 days) → behaviour stabilisation

This covers three critical layers of cognitive readiness:

Cognitive layer	Already covered by
Reasoning capability	MathTangle (Apply–Analyse–Evaluate)
Judgement in context	Practice Depth (esp. D3–D4)
Executive function / follow-through	Micro-task logging & consistency

NEP-2020 is not a curriculum upgrade only; it is a shift in how teaching decisions are made.
Such shifts cannot be sustained by skill acquisition alone.

Teachers need the **metacognitive readiness to notice when existing practices no longer serve learning, to question familiar routines, and to reframe instructional choices in response to new evidence**.

This readiness is not about intelligence or expertise.
It is about **cognitive flexibility, comfort with uncertainty, and willingness to unlearn before relearning**.

myelin does not treat metacognitive readiness as a separate assessment or judgement.
It is inferred through **how teachers engage with uncertainty (MathTangle patterns), how they reflect on practice (FP-3 depth), and how consistently they experiment over time (micro-task logging)**.

In the context of NEP-2020, this readiness determines not *whether* teachers believe in the reform, but *whether they can adapt their practice as the reform unfolds*.

MathTangle as a cognitive capacity signal

Using response patterns, not scores

1. MathTangle

MathTangle is not a test of correctness.
It is a probe of how a teacher engages with uncertainty, structure, and judgement.

So we extract **patterns of engagement**, not marks.

2. Data we capture

- selected option(s)
- time spent per item
- sequence/order of interaction
- whether answer was changed
- completion / skip behaviour

3. Cognitive pattern dimensions (CPDs)

These are **derived signals**, not scores.

CPD-1: Cognitive Engagement (Time-on-Task)

Signal	Interpretation
Very fast response across levels	Surface engagement / guessing
Moderate, stable time	Deliberate reasoning
Long time with abandonment	Cognitive overload or low confidence

CPD-2: Decomposition behaviour (Analyse proxy)

Signal	Interpretation
Linear click-through	Low decomposition
Pauses before option selection	Active analysis
Re-reading / hover behaviour (if available)	Sense-making attempt

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CPD-3: Evaluation & reconsideration

Signal	Interpretation
No answer changes	Fixed judgement (could be confident or rigid)
One revision	Reflective evaluation
Multiple revisions	Uncertainty or overthinking

CPD-4: Cognitive Persistence

Signal	Interpretation
Skips higher-level items	Avoidance of complexity
Completes all levels	Tolerance for ambiguity
Drops mid-way	Cognitive fatigue or resistance

CPD-5: Consistency Across Levels

Signal	Interpretation
High Apply, low Analyse/Evaluate engagement	Execution-heavy mindset
Stable engagement across levels	Balanced cognitive readiness
Drop only at Evaluate	Judgement under uncertainty is fragile

4. Cognitive readiness profiles (non-labelled, internal)

Using the CPDs above, **cluster**, not score.

Example internal profiles

Profile	Pattern Summary	What it means for change
Procedural Executor	Fast Apply, avoids Evaluate	Needs scaffolding before D3/D4
Careful Analyzer	Slow, stable, revises once	Good candidate for diagnostic practices
Reflective but Hesitant	Revises often, long time	Needs safety and affirmation
Adaptive Thinker	Engages deeply across levels	Ready for advanced micro-interventions

Avoidant Under Load	Drops at Analyse/Evaluate	Reduce cognitive load in tasks

These are **interpretive lenses**, not outputs.

5. How MathTangle patterns connect to existing stack

A. With Practice Depth

Observation	Interpretation
Low depth + low cognitive engagement	Capacity constraint
Low depth + high cognitive engagement	System/enablement constraint
Depth improves post micro-tasks + stable patterns	Transfer confirmed

B. With SET Archetypes

MathTangle **modulates** SET, it does not redefine it.

Computed **Normative NEP Orientation** that refers to the NEP Foundational Philosophy (FP) that a teacher most naturally aligns with when making value-based judgements about teaching before constraints, skills, or systems come into play.

Example: When your data shows a teacher has **high FP-2 orientation**, it means:

When faced with a pedagogical choice, this teacher is more likely to prioritise holistic, experiential considerations over other NEP lenses.

From Signal and data perspective **Normative NEP Orientation** is inferred from the relative frequency with which a respondent selects options mapped to a particular Foundational Philosophy (FP) across orientation questions, indicating the FP that most consistently guides their value-based pedagogical judgement.

Scenario 1: Teachers demonstrate a strong normative orientation towards FP-2 (Holistic & Experiential Learning)

Archetype: *Intent-High, Translation-Low*

- **Normative NEP Orientation:** High FP-2 orientation
  (teachers prioritise holistic and experiential considerations in value judgements)
- **Operational Practice Depth:** Low (D1–D2)
- **Adaptive / Metacognitive Capacity:** Strong (high Analyse/Evaluate engagement in MathTangle)

Interpretation:

Teachers have a clear value-based preference aligned with FP-2 and the cognitive capacity to adapt practice. The lack of deeper enactment is therefore more likely due to system or environmental constraints rather than a limitation in teacher capacity.

Scenario 2:

When FP-3-oriented teachers show strong evaluative thinking but only moderate practice depth, the constraint is likely safety to act and the not ability to reflect.

Archetype: *Reflection without safety*

- **Normative NEP Orientation:** Dominant FP-3 orientation
  (the teacher's value-based judgement prioritises reflection, professional judgement, and learning from experience)
- **Operational Practice Depth:** Mid-level (D2–D3)
  (reflection is present but has not yet translated into adaptive redesign of practice)
- **Adaptive / Metacognitive Capacity (MathTangle):**
  Frequent answer revisions and longer engagement at the *Evaluate* level

Interpretation:

The teacher demonstrates strong reflective thinking and evaluative judgement, but hesitates to act decisively on that thinking.
This suggests a lack of psychological or structural safety to experiment, rather than a limitation in cognitive capacity.
Interventions should therefore emphasise reassurance, protected experimentation, and supportive feedback, rather than additional skill training.

6. Summary table

Table: Cognitive Capacity Signal — MathTangle (Pattern-Based)

Dimension	Derived From	Used To Explain
Engagement	Time-on-task	Attention & effort
Decomposition	Interaction sequence	Analytical capacity
Evaluation	Answer revision	Judgement & reflection
Persistence	Completion across levels	Tolerance for complexity
Consistency	Level-wise behaviour	Readiness for D3/D4

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